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| **PI Name:** | Barmatz, Martin Ph.D. | | |
| **Project Title:** | Finite Size Scaling in He-3 | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FUNDAMENTAL PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FUNDAMENTAL PHYSICS--Low temperature condensed matter physics | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
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| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 91109 | **Congressional District:** | 29 |
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| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JPL |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Received NCE to 09/30/2007 per D. Strayer (1/07) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Cho, Hyung ( Jet Propulsion Laboratory )<br>Dantchev, Daniel ( Institute of Mechanics, Bulgarian Academy of Sciences )<br>Bonetti, Joseph ( Jet Propulsion Laboratory ) | | |
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| **Performance Goal Text:** | | | |

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| **Task Description:** | When a critical point system is spatially compact in one or more directions, its thermodynamic properties will deviate from bulk behavior as the critical point is approached. In particular, for the isothermal susceptibility, finite-size scaling theories predict a finite susceptibility maximum occurring at a temperature shifted from the bulk critical temperature. It has been assumed that standard finite-size scaling theories will completely describe the crossover behavior from the bulk to the finite-size regime, as well as the critical point Casimir effect. However, recent theoretical analyses have indicated that subleading long-range interactions can actually dominate the finite-size scaling divergences near a liquid-gas critical point. The 3He liquid-gas critical point was chosen as one of the systems to study this new prediction because it is a system devoid of impurities (except for a negligible concentration of 4He) and many sophisticated high precision experimental techniques have already been developed and tested to study this fluid. Efforts to test these new predictions in ground-based laboratories are limited due to a gravity-induced vertical density gradient associated with the divergence of the isothermal susceptibility that leads to a significant smearing of the transition.<br>The principal hypothesis in this proposal is that the isothermal susceptibility will suffer significant modifications near a liquid-vapor critical point when the measurements are performed in a geometry that enhances the surface-to-volume ratio of the system. An additional, closely related hypothesis is that the modifications of bulk critical behavior associated with surface effects are dependent in non-trivial ways on the properties of the surface with which the fluid is in contact. Recently, theoretical studies have demonstrated that subleading long-range interactions, exemplified by the van der Waals interaction, will influence finite-size effects near a liquid-gas critical point. A fundamental question is whether these long-range interactions will actually dominate the short-range Ising-like finite-size scaling behavior near the critical point in a real fluid system. This task will develop theoretical models for the isothermal susceptibility in the 2-D finite-size crossover region very near a liquid-gas critical point in both 1-g and microgravity environments. For the first time, the finite-size isothermal susceptibility will be measured near the 3He and 4He critical points to test the predictions of these models. The main experimental objective of this task is to measure the isothermal susceptibility along the critical isochore above the transition and along the coexistence curve below the transition to determine the critical amplitudes and exponents associated with this diverging thermodynamic quantity. A byproduct of these measurements will be the shape of the coexistence curve for this finite-size system. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The first test of the finite-size behavior of the susceptibility that can apply to all simple fluids will provide a major advancement in the field of condensed matter physics. |
| **Task Progress:** | We analyzed the behavior of the linear susceptibility in 3He and 4He thin films bounded by strongly adsorbing substrates as a function of the temperature T, the chemical potential μ, and gravity g. We have determined the behavior of the finite-size susceptibility in a microgravity environment and also included the effects due to gravity. We studied films with thicknesses spanning the range from L = 50 to L = 8000 fluid layers, where the thickness of the layer is taken to be equal to the average distance between the atoms of 3He or 4He. In our approach we took into account the direct substrate-substrate van der Waals interaction, the van der Waals interactions between the molecules of the fluid with the other molecules of the fluid as well as with the constituent elements of the substrate, the interaction between the plates generated by the fluctuations of the density of the fluid (i.e., the Casimir force) in mean-field approximation, and the effects due to the gravity field of the Earth. Thin films with thickness up to L = 4000 are strongly influenced by the substrate-fluid van der Waals interaction while gravity effects start to be visible from L = 4000 and become dominant for thick films represented with L = 8000.<br>Within our model, we have calculated the order parameter (density) profiles for which the susceptibility reaches its maximum. We have also determined the relevancy of gravity as a function of the thickness L of the films: for thin films it is negligible, for thick ones it is not. Films with L = 1000, 2000 or 4000 liquid layers are, in this sense, thin films, while film with, say, L = 8000 layers, is already a thick film. We find that both the van der Waals interactions, as well as the presence of gravity, reduce the magnitude of the susceptibility. This is due to the ordering effect of the van der Waals and gravity field: for L = 1000, 2000 or 4000, when the van der Waals interactions are the important factor which causes the effect, this reduction is approximately given by a factor of 2, while for L = 8000 this reduction is already 3 times and is mainly due to the influence of the gravity. We believe that these curves shall resemble the ones one shall obtain in experiments with liquid van der Waals films.<br><br>The primary experimental goal of this experiment was to measure the linear susceptibility in 3He and 4He near their liquid-gas critical point in confined geometries. The desired confining finite-size dimension was on the order of 1 micron. The linear susceptibility was measured with the sweep electrostrictive technique. An applied electric field across a capacitor in a dielectric fluid causes a change in pressure between the fluid in the capacitor gap and the surrounding medium. This change in the pressure leads to a change in density in the dielectric fluid between the plates of the capacitor. This change in density can be determined by using the Claussius-Mossotti relation after measuring the change in capacitance of the capacitive density sensor. Our first attempt was to have an outside company fabricate a cell from bonded silicon wafers. There were four gap sizes, 0.5, 1, 2, and 4 microns.<br><br>Measurements on this cell proved to be cumbersome. Although the wafers were made with low-doped silicon, the resistivity of the wafers at room temperature, and even at liquid nitrogen temperature, was low enough that it inhibited the measurement of the capacitance. The reason for this result was that the capacitors were essentially shorted out by the low resistance of silicon. The electrical resistance of silicon increases exponentially with decreasing temperature, so at liquid helium temperatures we were able to measure a capacitance. However, the capacitance values were always incorrect given the area and the gap we assumed. The capacitance values were consistently lower than we expected by roughly a factor of 10. The areas are well defined, so the only explanation was that gap sizes were bigger than advertised by a factor of 10.<br><br>We decided to build our own cell using sapphire discs. Using sapphire had a few advantages. One, it is an electrical insulator at room temperature, allowing us to make room temp measurements prior to cooling the cell to cryogenic temperatures. Two, the sapphire discs were bolted together with ceramic bolts, allowing for disassembly and repair if needed. Measurements using the sapphire cell were much easier and more reasonable. Just as in the silicon cell, the gap sizes were 0.5, 1, 2 and 4 microns. Room temperature measurements on the 2 and 4-micron gap capacitors gave reasonable values that agreed with calculation to within 10%. However the 1 and 0.5 micron gap capacitors had values that were about a factor of 1.5 to 2 lower than expected. The cell was disassembled, cleaned, and reassembled several times in the microdevices lab clean room. Each time, we saw the same result. Even though the smaller of the four gaps had the wrong capacitance we decided to cool the cell to liquid nitrogen temperatures. This was successfully done, and |

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| | the cell and capacitors held up just fine. Unfortunately, due to time and funding constraints we were not able to cool to liquid helium temperatures or perform the final measurements to extract the susceptibility. We have confidence that this experimental approach is viable and should be successful upon further evaluation. This experimental system should be able to test the theoretical predictions obtained in this task, both on earth in a 1-g environment as well as in a future microgravity experiment. |
| **Bibliography Type:** | Description: (Last Updated: 08/04/2008) |
| **Articles in Peer-reviewed Journals** | Dantchev D, Rudnick J, Barmatz M. "Finite-size effects on the behavior of the susceptibility in van der Waals films bounded by strongly absorbing substrates." Phys Rev E Stat Nonlin Soft Matter Phys. 2007 Jan;75(1 Pt 1):011121. Epub 2007 Jan 23. PMID: 17358124 , Jan-2007 |